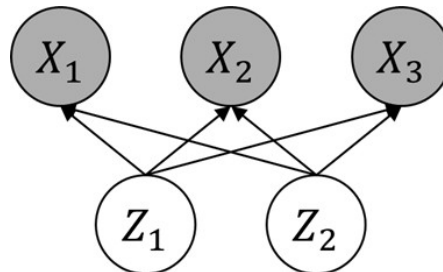


Chapter 2 (Part II): Random Variables

Conditional distributions and Bayesian networks: Express the joint probability density function of the following Bayesian network, and provide example data generated from this structure:



Chapter 3: Statistical Estimation and Hypothesis Testing

Likelihood: Formulate likelihood functions given probabilistic models

Maximum likelihood estimation: Derive maximum likelihood estimators (MLEs) for Gaussian distributions, and express MLEs as minimizers of the Kullback-Leibler (KL)-divergence between the model density and the empirical data density.

Statistical hypothesis test: Understand the Neyman-Pearson framework and that likelihood ratio-based tests are optimal under this framework.

Two-sample comparison: Compare two Gaussian distributions using samples:

$$H_0: N(\mu_X, \sigma_X^2) = N(\mu_Y, \sigma_Y^2) \quad \text{vs.} \quad H_1: N(\mu_X, \sigma_X^2) \neq N(\mu_Y, \sigma_Y^2)$$

The trinity of hypothesis tests: Understand that the (generalized) likelihood ratio test, Wald test, and score test intuitively compare vertical, horizontal, and slope differences between the model and data likelihoods

